

123456

SIMPLE HOBBY CONTROLLER – TWO READABLE SCHEMATIC PAGES

Power + dual wheel driver

File: simple_01_power_motors.kicad_sch

ESP32 + servos + UART

File: simple_02_esp_servos_uart.kicad_sch

Harrison

Sheet: /

File: pickleball_robot_simple.kicad_sch

Title: **Pickleball Robot – Simple Hobby Controller**

Size: A4

Date: 2026-07-12

KiCad E.D.A. 10.0.4

Rev: **H1-SIMPLE**

Id: 1/3

A

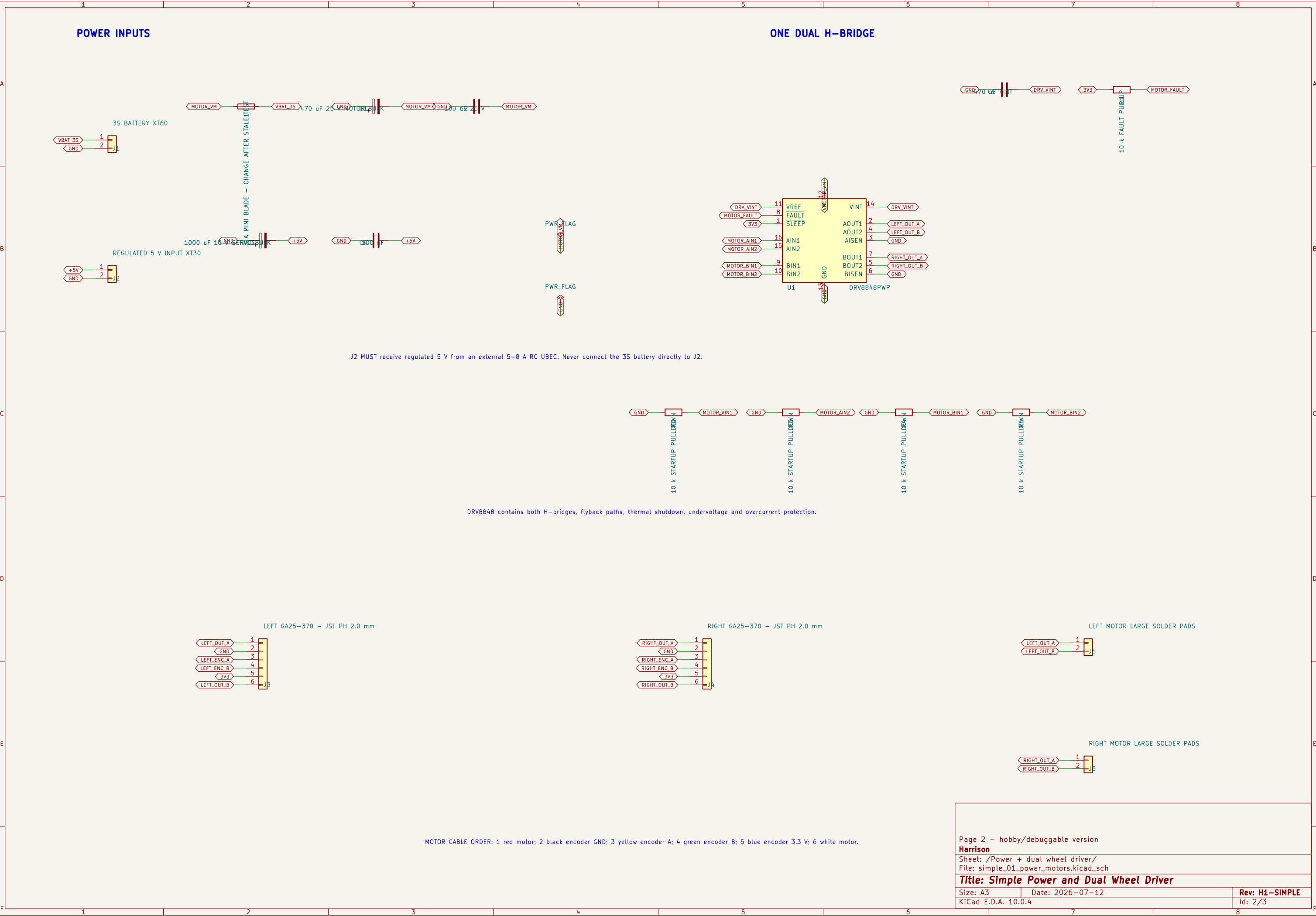
B

C

D

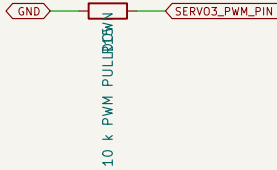
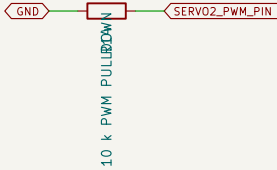
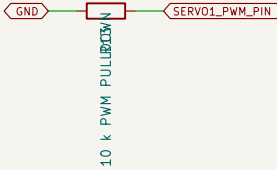
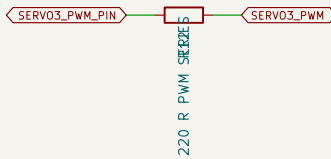
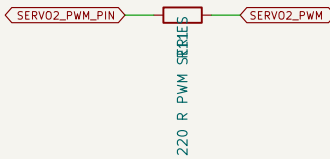
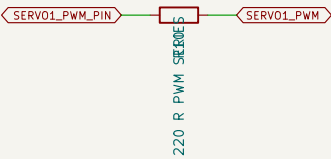
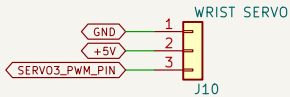
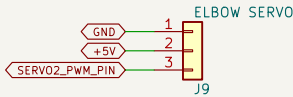
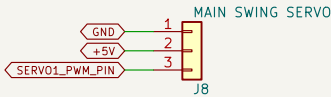
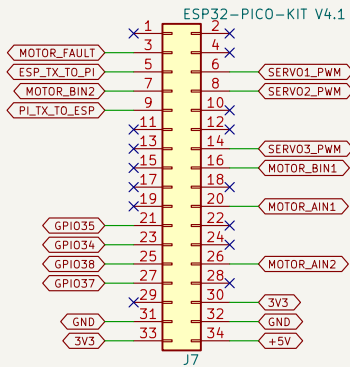
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Open a page from the hierarchy panel. Global labels join nets between the two pages.



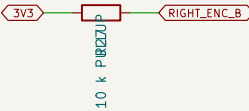
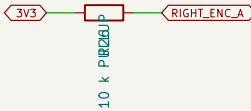
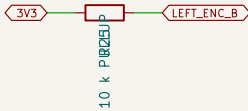
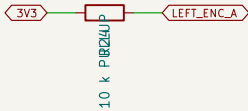
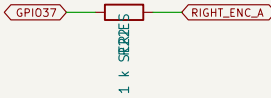
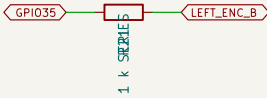
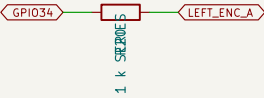
ESP32-PICO-KIT V4.1 CARRIER

THREE INDEPENDENT 5 V SERVOS

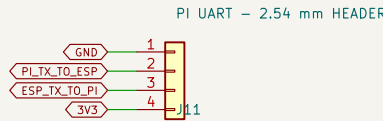


Servo headers: pin 1 GND, pin 2 regulated +5 V, pin 3 separate PWM. Each servo moves independently.

ENCODER INPUTS



RASPBERRY PI 3.3 V UART



J11: 1 GND; 2 PI TX -> ESP GPIO18 RX; 3 ESP GPIO19 TX -> PI RX; 4 is 3.3 V reference only.

Motor encoder outputs are assumed 3.3 V because their blue supply wire is powered from 3.3 V.

DO NOT power the Raspberry PI from J11. Power the PI from its own proper 5 V supply/USB-C regulator.

Page 3 - hobby/debuggable version

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Sheet: /ESP32 + servos + UART/
File: simple_02_esp_servos_uart.kicad_sch

Title: ESP32, Three Servos, Encoders, and PI UART

Size: A3 Date: 2026-07-12

KiCad E.D.A. 10.0.4

Rev: H1-SIMPLE

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